

Issues Identified — Task Manager Debug Capstone

Variables and Operators:

- Global variable declared without var/let/const (implicit global leak)
- Multiple var declarations replaced with let/const for proper block scoping
- Loose equality (==) used instead of strict equality (===) in comparisons
- Assignment operator (=) mistakenly used inside a conditional instead of ===

Control Flow:

- Off-by-one error in for loop (<= instead of <), causing undefined array access
- Infinite loop in while loop due to missing increment (i++)
- Manual for loops replaced with for-of loops and array methods where appropriate

Functions:

- findTaskByTitle was missing its title parameter entirely
- Recursive function (countCompletedTasks) had no base case, causing stack overflow
- No parameter validation existed on core functions (addTask, updateTaskPriority)

OOP and Classes:

- Task class was missing an id property
- Task class was missing a toggleCompleted method
- SubTask class was missing its super() call, breaking inheritance entirely
- TaskManager object was missing functional methods (filter/map based)

Modern JavaScript:

- No destructuring used anywhere in the original code
- No spread/rest operators used; manual loops used to merge arrays instead
- String concatenation used instead of template literals

DOM and Events:

- Incorrect selector usage mixing ID and class syntax
- No null checks before attaching event listeners
- No event delegation; individual listeners would have been required per task
- Script initialization ran before DOMContentLoaded, risking timing bugs

Storage and JSON:

- localStorage calls never used JSON.stringify/JSON.parse, corrupting saved data

- No error handling around storage or parsing failures
- Reloaded tasks lost their class methods since `JSON.parse` only returns plain objects — fixed by rehydrating parsed data back into Task instances

Error Handling:

- No try-catch blocks existed anywhere in the original codebase
- Functions failed silently or crashed instead of surfacing meaningful errors

Testing:

- Original test file had only 2 shallow tests and no error/edge case coverage
- Expanded to 20+ tests covering classes, inheritance, recursion, array operations, destructuring, error handling, and edge cases